The velocity dispersions of thin DESI bins

The halo-model physics of σ_v, and a study that gives it a per-bin slope and correlated scatter

November 2026

1 Summary

The setting. The project forecasts DESI galaxy clustering in thin redshift bins – angular power spectra, joint with LSST Y1 cosmic shear – with Fisher matrices: the forecast's error bars come from the derivatives of the predicted data at one fiducial point. That machinery, including the figure of merit FoM $= [\det C]^{-1/2}$ for a parameter pair (the inverse area of its error ellipse, with C the pair's forecast covariance), is built step by step in the machinery note, `notes/fisher_machinery`; this note needs only its vocabulary.

The question. The forecast blurs each thin bin's redshift distribution by a Gaussian whose width is set by one velocity dispersion σ_v per dispersion group (a set of neighbouring bins that share one σ_v value): seven numbers, 371 to 289 km/s, each shared by 8 to 25 bins (Section 3). The galaxy population changes across a group – host halos, satellite fractions and selection cuts all drift with redshift – so the true dispersion cannot be exactly flat. Two questions follow. Is the one-σ_v-per-group model biased when the truth varies from bin to bin? And what does per-bin freedom cost in FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$?

The physics. Sections 4.1–4.7 build σ_v from first principles, step by step with worked numbers. A galaxy's line-of-sight velocity is the sum of two independent parts: the bulk motion of its host halo, set by the large-scale flow of matter ($\sigma_{\rm lin} \approx$ 260–310 km/s at the fiducial cosmology, Section 4.4), and its motion inside the halo, set by the halo's depth, the satellite fraction and the velocity biases of the sample ($\sigma_{\rm HOD} \approx$ 110–210 km/s, Sections 4.1–4.3). The variances add, $\sigma_v^2 = \sigma_{\rm lin}^2 + \sigma_{\rm HOD}^2$ (Section 4.5), and the halo-model prediction for the DESI LRGs, $\sigma_{\rm HOD} \approx 170$ km/s at $z = 0.5$, agrees with the value implied by the project's fiducials to about 10%. Only the halo part can change from bin to bin: the linear part is fixed by cosmology and smooth in redshift. Its share of the variance, $w = \sigma_{\rm HOD}^2/\sigma_v^2$, is 0.27–0.33 for BGS and LRG and 0.14–0.17 for ELG, so a given fractional wiggle of the halo part reaches σ_v suppressed by w.

The model. The per-bin model (Section 6) keeps the fiducial and the group-level prior unchanged and adds, on the halo part only, a free slope in redshift per group and a correlated per-bin residual with the exponential-kernel prior of the linear bias, projected so that it charges neither the level nor the slope. To first order the change of any bin's dispersion is $\delta\sigma_{v,i} = w_i\,\sigma_{v,g}\,[\,s_g(\bar z_i - \bar z_g) + \varepsilon_i\,]$, so the study needs only per-bin σ_v derivative columns, no new physics in the likelihood. The priors (slope widths 0.5/1.0/2.0 per unit z, BGS 2.0; residual A = 10% of $\sigma_{\rm HOD}$ with correlation length $L = 0.2$; residual B = 20%, BGS 30%, with $L = 0.05$) each follow from a number derived in Sections 4 and 5.

parameter	meaning	fiducial	prior
$10^9 A_s$	primordial curvature amplitude	2.1	flat
n_s	scalar spectral index	0.96605	flat
H_0	Hubble constant [km/s/Mpc]	67.32	flat
Ω_b	baryon density today	0.04	flat
Ω_m	matter density today	0.3	flat
w_0	dark-energy equation of state today	-1	flat
w_a	its slope, $w(a) = w_0 + w_a(1-a)$	0	flat, as $w_0 + w_a$
$\sum m_\nu$	neutrino mass sum [eV]	0.06	fixed in the Fisher
τ	optical depth to reionization	0.0697186	fixed

Table 1: The fiducial cosmology: the evaluate `override` of the paper yamls. The yamls' `ref` entries (chain starting points) differ from it and are not used.

The study. Before any figure of merit, an accuracy test fits today's model to truths that vary bin by bin – the linear flow's own redshift dependence, which needs no prior choice, and random draws from the per-bin priors – and scores $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ for (w_0, w_a), Ω_m and A_s. Then five variants, from today's model (V0) to every per-bin σ_v free (V4), give the cost of the freedom (Section 7). The results (Section 8): under $T_{\rm lin}$ alone – the shape cosmology fixes, no prior involved – today's model moves the (w_0, w_a) fit by up to 0.32 of its error ellipse on $\Delta z = 0.02$ at $k_{\rm max} = 0.2\,h$/Mpc and up to 0.10 on $\Delta d_{\rm com} = 62$ Mpc. One free halo-part slope per group absorbs a third of that shift (0.32 → 0.21) and keeps $\eta \le 1.016$ under the truth that adds random slopes and residual-B wiggles, at a cost of at most 3.4% of either FoM; every per-bin σ_v free costs 5–11% on the three thinner layouts at $k_{\rm max} = 0.2\,h$/Mpc. The open decisions, including whether to build the linear part's redshift dependence into the fiducial itself, are in the appendix "Open issues".

2 Fiducial model and parameter values

This section freezes the model and the numbers this note is built on, so the note does not depend on files that may drift.

The cosmology. The fiducial point is the evaluate `override` block of the paper yamls `EXAMPLE_EVALUATE1–4.yaml`, the same in all four (Table 1). At this point $\sigma_8 = 0.8277$. Every linear-theory number of this note comes from one CAMB linear $P(k)$ at this cosmology (the generators in `plots/`); at it, the one-dimensional linear displacement (defined in Section 4.4) is $\sigma_{d,0} = 8.86$ Mpc $= 5.96\,$Mpc$/h$.

The layouts and the dispersion groups. The four layouts slice the DESI samples into thin bins: $\Delta d_{\rm com} = 62$ Mpc (76 bins of equal comoving width) and $\Delta z = 0.02$, 0.05, 0.2 (90, 36, 10 bins). Bin i has mean redshift $\bar{z}_i$. The samples are DESI's three tracer types: BGS, the Bright Galaxy Survey sample, a flux-limited low-redshift sample; LRGs, luminous red galaxies, massive and strongly clustered; and ELGs, emission-line galaxies, star forming and hosted by much lighter halos. Their six bias groups (BGS, LRG1, LRG2, LRG3, ELG1, ELG2) become seven dispersion groups g by splitting BGS at $\bar{z} = 0.255$, where the source

group	$\bar{z}_g$	σ_v [km/s]	exploration width	paper width
BGS0 ($\bar{z}_i < 0.255$)	0.18	371	145	870
BGS1 ($\bar{z}_i \geq 0.255$)	0.33	358	105	630
LRG1	0.50	357	70	420
LRG2	0.70	348	50	300
LRG3	0.95	343	50	300
ELG1	0.95	304	40	240
ELG2	1.35	289	30	180

Table 2: The seven dispersion groups: the group mean redshift on $\Delta z = 0.02$, the fiducial dispersion, and the widths of the Gaussian prior in the exploration files and in the paper model (both in km/s).

forecast's measured dispersion curve steps. (The source forecast is the external DESI forecast whose survey inputs – redshift distributions, number densities, the measured dispersions below – the project imported.) On $\Delta z = 0.02$, the layout of every per-bin figure of this note, the groups hold bins 1–8, 9–15, 16–25, 26–35, 36–50, 51–65 and 66–90, with group mean redshifts $\bar{z}_g = 0.18$, 0.33, 0.50, 0.70, 0.95, 0.95 and 1.35; the other layouts' $\bar{z}_g$ differ by less than 0.05. In each group the first bin's `DESI_SIGV` parameter carries the prior and the other bins are tied to it by value.

The dispersions and their priors. Table 2 gives each group's fiducial σ_v and the widths of its Gaussian prior: the exploration width (five times the median per-bin error bar of the source forecast's measurement) and the paper width, six times larger. The yaml priors are truncated at $\sigma_v = 0$; at the paper width the BGS0 truncation is severe (a Gaussian truncated at zero has mean $\mu + \sigma\,\phi(\mu/\sigma)/\Phi(\mu/\sigma)$, with ϕ and Φ the standard normal density and distribution: here $371 + 870 \times 0.364/0.665 \approx 850\,\mathrm{km/s}$, not the fiducial 371), while the Fisher matrices treat every σ_v prior as an untruncated Gaussian of the quoted width centered on the fiducial.

The blur in the likelihood. Each bin's raw redshift distribution $n_i(z)$ is convolved with a Gaussian of width $\sigma_{z,i} = (1 + \bar{z}_i)\,\sigma_v/c$ in redshift (the Doppler relation of Section 3, applied to the rms velocity; `set_lens_related` in `likelihood/desi_cl.py`, $c = 299792.458\,\mathrm{km/s}$), at every sampled point. The construction, the grid and the overlap consequences are derived in the cross-correlation note, `notes/cross_correlations`; the degeneracy of σ_v with the radial counterterm α_2, and why the paper model fixes $\alpha_2 = 0$ and widens the σ_v priors sixfold in exchange, are the Fingers-of-God note, `notes/fog_parameter_degeneracy_study`.

Sources of the halo-model inputs. The virial calibration is Evrard et al. [3]. The DESI halo-occupation fits are the One-Percent LRG analysis [6] and ELG analysis [7] on AbacusSummit halos found by CompaSO [8], with the velocity-bias definitions of AbacusHOD [5]. The stochastic-term scale is Maus et al. [14]. Every number quoted from these papers was read from their published tables; numbers this note computes from them are marked as estimates.

3 What σ_v is

Redshift is not distance. The forecast places each galaxy at its measured redshift. That redshift has two parts: the cosmological redshift, set by the galaxy's distance, and a Doppler shift from the galaxy's own line-of-sight velocity v, its peculiar velocity. The two multiply, $1 + z_{\rm obs} = (1 + z)(1 + v/c)$, so for $v \ll c$

$$z_{\rm obs} - z \;\simeq\; (1+z)\,\frac{v}{c}, \qquad \delta\chi \;=\; \frac{c\,(z_{\rm obs} - z)}{H(z)} \;=\; \frac{(1+z)\,v}{H(z)}, \tag{1}$$

with χ the comoving distance and $H(z)$ the expansion rate; the second relation follows from $\chi(z) = \int_0^z c\,dz'/H(z')$, so a small redshift error δz moves the inferred distance by $c\,\delta z/H(z)$. A dispersion σ_v of these velocities therefore blurs each bin's map of the matter along the line of sight by $\sigma_\chi = (1+z)\,\sigma_v/H(z)$: at $z = 0.5$, where $H = 88\,$km/s/Mpc at the fiducial cosmology, $\sigma_\chi = 1.5 \times 357/88 \approx 6$ Mpc, and the other groups give 5–6 Mpc. This is why *thin* bins care: 6 Mpc is a tenth of a 62 Mpc bin, so the blur smears each bin's window and moves galaxies near an edge into the neighbouring bin, while a $\Delta z = 0.2$ bin, several hundred Mpc deep, hardly notices. The effect is the Fingers-of-God blur [1], named for the way dense regions, stretched along the line of sight by their internal velocities, point at the observer in a redshift map. The coherent part of the same velocities – matter streaming onto growing structures – is the Kaiser distortion [2], which the model computes separately; Section 4.7 returns to the boundary between the two.

One number per group. σ_v is the *one-point* rms line-of-sight peculiar velocity of a bin's galaxies: the standard deviation of one galaxy's velocity, not of a velocity difference between two galaxies (Section 4.6 makes the distinction precise). The source forecast measured it from its simulated galaxies as a function of redshift, with an error bar per bin, and the project read one value per group off that measurement. The measurement's per-bin figure is not available now; the repository keeps the seven group values and the per-bin error bars (29/21/14/10/10/8/6 km/s).

The question of this note. What physics sets 371 to 289 km/s, and how much can the true dispersion vary between the bins of one group? The next section builds the answer from halos up.

4 The halo model of galaxy velocities, step by step

The halo model describes the galaxy distribution in two layers: dark matter collects into gravitationally bound clumps (halos), and galaxies occupy halos. A galaxy's velocity then also has two layers,

$$v \;=\; v_{\rm halo} \;+\; v_{\rm int}\,: \tag{2}$$

the bulk velocity of its host halo, carried by the large-scale flow, and its own motion inside the halo. Figure 1 draws the decomposition and the blur it produces across a thin bin's edge. The four steps below compute the dispersion of each part and put them together: how fast matter moves inside a halo of mass M (Section 4.1); how galaxies move compared with that matter (Section 4.2); the average over the galaxies of a sample (Section 4.3); and the halos' own motion (Section 4.4).

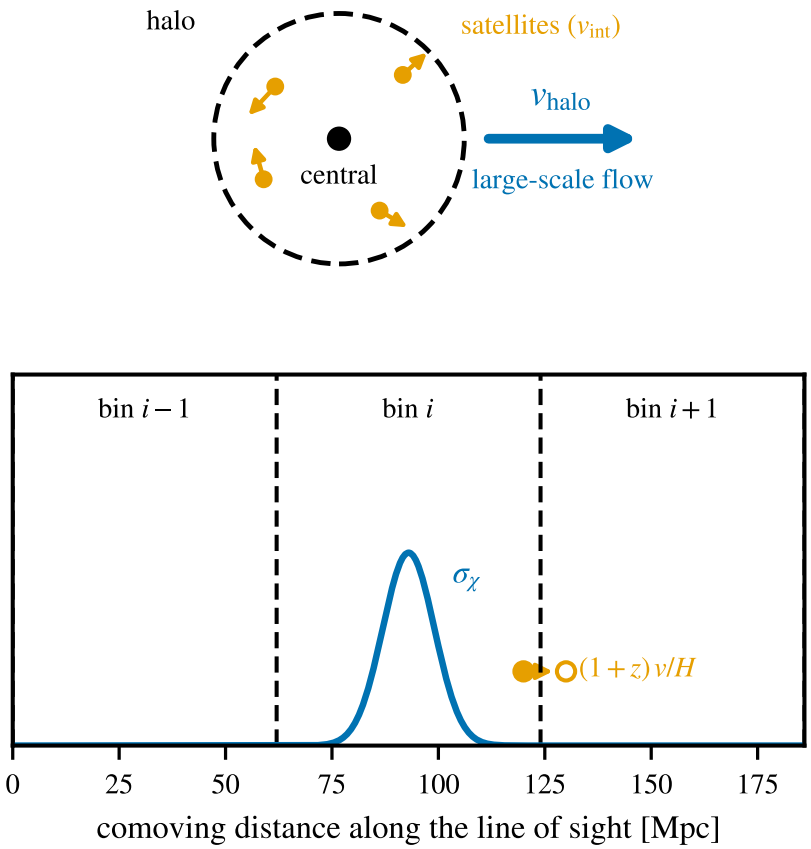


Figure 1: The two parts of a galaxy's velocity (Eq. (2)), and the blur they produce. Top: a halo drifts as a whole with the large-scale flow ($v_{\rm halo}$, Section 4.4), while its central sits near the center and its satellites orbit inside it ($v_{\rm int}$, Sections 4.1–4.3). Bottom: the line-of-sight velocity shifts a galaxy's apparent comoving position by $(1+z)\,v/H(z)$ (Eq. (1); filled circle the true position, open circle the apparent one), so a galaxy near a bin edge is counted in the neighbouring bin and the bin's galaxies smear by σ_χ; the blur and the 62 Mpc bin width are drawn to scale at $z = 0.5$.

4.1 Step 1: how fast matter moves inside a halo

Virial equilibrium, in plain words. A halo is a region that has collapsed under its own gravity and settled into a steady state: its particles orbit in the common potential well, falling in and swinging out, with no net contraction or expansion. In such a state the virial theorem ties the kinetic energy T to the gravitational potential energy W, $2T = -W$ up to boundary terms, so the typical squared velocity is set by the well's depth: $\sigma^2 \sim GM/R$, with M the halo's mass and R its radius. Deeper well, faster orbits.

The radius is not free. A halo's "edge" is a convention: the radius inside which the mean density exceeds a chosen threshold Δ times a reference density. The common choice M_{200c}

uses $\Delta = 200$ times the *critical* density $\rho_c(z) = 3H^2(z)/(8\pi G)$:

$$M_{200c} \;=\; \frac{4\pi}{3}\,200\,\rho_c(z)\,R_{200c}^3 \;=\; \frac{100\,H^2(z)\,R_{200c}^3}{G}. \tag{3}$$

Solving for R_{200c} and inserting it into $\sigma^2 \sim GM/R$ gives the scaling that organizes everything below:

$$\sigma^2 \;\propto\; GM^{2/3}H^{2/3}(z) \qquad \Longrightarrow \qquad \sigma_{\rm halo} \;\propto\; \big[M\,H(z)\big]^{1/3}. \tag{4}$$

The mass enters as $M^{1/3}$: a thousand times more massive halo is only ten times faster inside. And at fixed mass a halo at higher redshift is faster, because the universe was denser, so the same mass fits inside a smaller, deeper well.

A worked normalization. The simplest concrete profile is the singular isothermal sphere, a halo with a flat circular-velocity curve $V_c^2 = GM(<r)/r$, with $M(<r)$ the mass enclosed within r; a standard property of this profile is that its one-dimensional velocity dispersion is $\sigma = V_c/\sqrt{2}$, so $\sigma^2 = GM_{200c}/(2R_{200c})$. For $M_{200c} = 10^{15}\,M_\odot$ at $h(z) \equiv H(z)/(100\,\mathrm{km/s/Mpc}) = 1$, Eq. (3) gives $R_{200c} = 1.63$ Mpc and $\sigma = 1150\,\mathrm{km/s}$ (with $G = 4.30\times10^{-9}\,\mathrm{Mpc\,(km/s)^2}/M_\odot$, so that masses in $M_\odot$ and lengths in Mpc give velocities in km/s).

The simulation calibration. Evrard et al. [3] measured this relation in an ensemble of N-body simulations. They define $\sigma_{\rm halo}$ as the one-dimensional dispersion of the dark-matter particle velocities, averaged over the three components, $\sigma^2_{\rm halo} = (3N_p)^{-1}\sum_i\sum_{j=1}^3|v_{i,j}-\bar v_j|^2$, for halos of mass M_{200c}, and fit

$$\sigma_{\rm halo}(M,z) \;=\; 1082.9\pm4.0\,\mathrm{km/s}\left[\frac{h(z)\,M_{200c}}{10^{15}\,M_\odot}\right]^{0.3361\pm0.0026}, \tag{5}$$

with a log-normal scatter of 4.3% at fixed mass. Three readings of Eq. (5). First, the slope is the virial 1/3 of Eq. (4) to one percent, across five decades of mass. Second, the normalization sits 6% below the isothermal-sphere estimate above – the dimensional argument is honest. Third, the $h(z)$ factor is exactly the $[MH(z)]^{1/3}$ scaling: with the mass in $M_\odot/h$, as the DESI papers quote it, $h(z)\,M[M_\odot] = E(z)\,M[M_\odot/h]$, where $E(z) = H(z)/H_0$.

Worked numbers at the DESI masses. Figure 2 draws Eq. (5) at the group redshifts, with the host masses derived in the next two steps marked. At $z = 0.5$ ($E = 1.31$): a halo of $10^{13.3}\,M_\odot/h$ – the two-mass toy's stand-in for the LRG centrals' hosts – has $\sigma_{\rm halo} = 318\,\mathrm{km/s}$, and one of $10^{13.9}\,M_\odot/h$ – the satellites' hosts – has 506 km/s. At the mean host masses of the full calculation of Section 4.3, $10^{13.04}$ and $10^{13.82}\,M_\odot/h$, the values are 260 and 476 km/s. At $z = 0.95$ the mean ELG host of $10^{11.78}\,M_\odot/h$ gives only 107 km/s: ELGs live in halos about twenty times lighter than the LRG centrals' hosts and a hundred times lighter than the LRG satellites' hosts, and that, through $M^{1/3}$, is why their dispersions are the smallest of the survey.

Mass definitions differ, and it matters at the few-percent level. The DESI fits quote CompaSO halo masses: the total mass of the particles competitively assigned to the halo [7, 8], whose outer boundary is a spherical-overdensity threshold $\Delta_{L1}(z) = (200/18\pi^2)\,\Delta_{\rm BN}(z)$ relative to critical, with $\Delta_{\rm BN}$ the spherical-collapse fit of Bryan and Norman [9] ($18\pi^2 + 82x -$

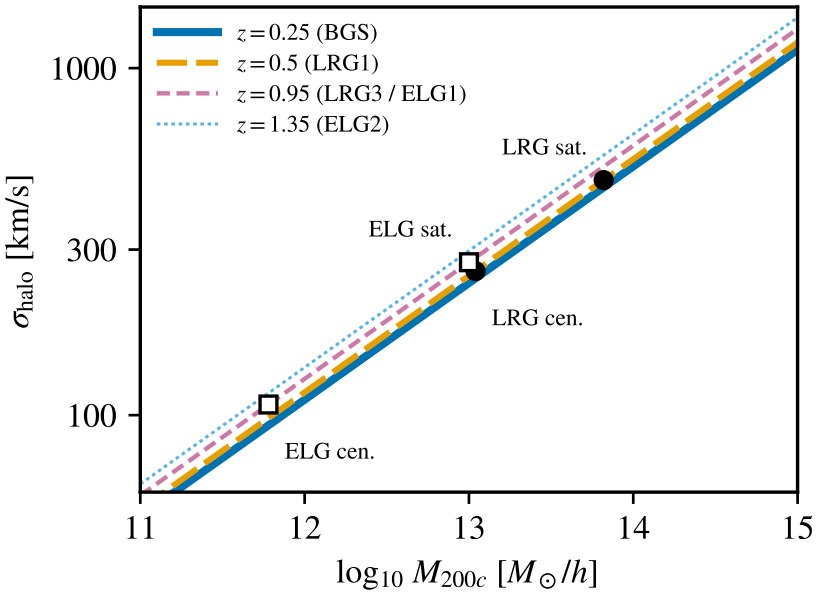


Figure 2: The virial velocity $\sigma_{\rm halo}(M_{200c}, z)$ of Eq. (5) at the mean redshifts of four dispersion groups. Filled circles: the mean host masses of the DESI LRG centrals and satellites at $z = 0.5$ (Sections 4.2 and 4.3); open squares: the ELG centrals and satellites at $z = 0.95$. The singular-isothermal-sphere estimate at $z = 0.5$ is not drawn; it runs 6% above the $z = 0.5$ curve.

$39x^2$, $x = \Omega_m(z) - 1$). At $z = 0.5$ that threshold is $155\,\rho_c$, looser than Evrard's $200\,\rho_c$, so a CompaSO sphere reaches further out and holds more mass: for an NFW halo of concentration 5–7, $M_{L1}/M_{200c} \simeq 1.07$–$1.08$ at $z = 0.5$, falling to 1.04 at $z = 0.95$ and 1.03 at $z = 1.35$ as $\Delta_{L1} \to 200\,\rho_c$ (an estimate from the profile alone; the competitive assignment can only remove mass). Plugging a raw CompaSO mass into Eq. (5) therefore overestimates $\sigma_{\rm halo}$ by about $(M_{L1}/M_{200c})^{1/3} - 1 \approx 2$–$3\%$ at $z = 0.5$ and by about 1% at the ELG redshifts. This note's numbers absorb no such conversion: a 2–3% bias on $\sigma_{\rm halo}$ is far below the velocity-bias and satellite-fraction uncertainties of the next steps, but it belongs on any future precision budget.

4.2 Step 2: galaxies in halos – occupation and velocity bias

The HOD, in plain words. A halo occupation distribution (HOD) gives, for a halo of mass M, the mean number of galaxies of the sample it hosts, split into a *central* – at most one, sitting near the halo's center – and *satellites*, orbiting inside it. The LRG fits use the standard five-parameter form of Zheng et al. [4], as AbacusHOD writes it [5]:

$$\langle N_c\rangle(M) = \tfrac{1}{2}\,\mathrm{erfc}\left[\frac{\log_{10}(M_{\rm cut}/M)}{\sqrt{2}\,\sigma}\right], \qquad \langle N_s\rangle(M) = \left[\frac{M - \kappa M_{\rm cut}}{M_1}\right]^{\alpha} \langle N_c\rangle(M). \tag{6}$$

Read it as follows: erfc is the complementary error function, falling smoothly from 2 to 0, so $\langle N_c\rangle$ turns on around the mass scale $M_{\rm cut}$ (where it equals 1/2) over a width σ in dex (decimal exponents: 0.3 dex is a factor $10^{0.3} \approx 2$ in mass); satellites appear above $\kappa M_{\rm cut}$ and their number grows as a power law, reaching about one per halo at M_1, typically ten times $M_{\rm cut}$.

	$\xi(r_p, r_\pi) + n(z)$ fits		$w_p + n(z)$ fits	
LRG	$0.4 < z < 0.6$	$0.6 < z < 0.8$	$0.8 < z < 0.95$	$0.95 < z < 1.1$
$\log_{10} M_{\rm cut}$	12.79	12.64	12.89	12.68
$\log_{10} M_1$	13.88	13.71	13.96	13.60
$f_{\rm sat}$	0.106 ± 0.012	0.136 ± 0.011	0.110 ± 0.014	0.151 ± 0.045
$\log_{10} \overline{M}_h$	13.40 ± 0.02	13.24 ± 0.02	13.29 ± 0.02	13.00 ± 0.03
α_c	$0.33^{+0.05}_{-0.07}$	$0.19^{+0.06}_{-0.09}$	–	–
α_s	0.80 ± 0.07	$0.95^{+0.07}_{-0.06}$	–	–
completeness $f_{\rm ic}$	0.70	0.62	0.92	0.19
linear bias b	1.93	2.08	2.31	2.13

Table 3: The DESI One-Percent LRG HOD fits [6]: the two $z < 0.8$ bins fitted to the redshift-space correlation function $\xi(r_p, r_\pi)$ (r_p, r_π the pair separations across and along the line of sight), which constrains the velocity biases, and the two $z > 0.8$ bins fitted to the projected clustering w_p, which is blind to velocities. Masses in $M_\odot/h$; $\overline{M}_h$ is the mean host mass per galaxy. The authors flag that fixed-redshift snapshots bias the $\overline{M}_h$ comparison between bins.

The DESI LRG fits. Table 3 collects what this note needs from the One-Percent fits [6]. The satellite fraction $f_{\rm sat}$ – the fraction of the sample's galaxies that are satellites – rises from 9–11% at $z \simeq 0.5$ to 15% in the last bin (w_p series: 8.9, 10.4, 11.0, 15.1%), while the mean host mass falls from $\log_{10} \overline{M}_h = 13.42$ to 13.00 (w_p series). The last bin is special. The sample's completeness $f_{\rm ic}$ (the overall factor by which the fitted occupation is scaled down, absorbing targets the selection misses) collapses from 0.92 to 0.19, and the linear bias b (the factor by which the sample's density contrast exceeds the matter's on large scales) drops from 2.31 to 2.13, all within $\Delta z \approx 0.15$: the selection, not the halos, changes there. Section 5 builds the "selection edge" residual prior on exactly this behavior.

Velocity bias: the definitions. Galaxies need not move exactly like the dark matter of their halos, and the fits parameterize the difference [5]:

- **Centrals.** The central sits on the halo's central subhalo and gets an extra Gaussian line-of-sight velocity, $v_{{\rm cen},z} = v_{{\rm halo},z} + \alpha_c\, \delta v(\sigma_{\rm LOS})$, where δv is a Gaussian draw of width equal to the halo's own particle dispersion. So α_c is the central's residual dispersion *in units of* $\sigma_{\rm halo}$; $\alpha_c = 0$ means the central rides the halo exactly.
- **Satellites.** Each satellite rides a halo particle with its velocity rescaled about the halo's bulk motion, $v_{{\rm sat},z} = v_{{\rm halo},z} + \alpha_s\, (v_{{\rm p},z} - v_{{\rm halo},z})$. So α_s multiplies the satellite dispersion relative to $\sigma_{\rm halo}$; $\alpha_s = 1$ means satellites move like the dark matter.
- **ELGs.** The ELG fits [7] use a different but equivalent satellite convention: satellite velocities are drawn from a Gaussian about the halo's velocity with width $f_{\sigma_v}\, \sigma_{\rm halo}$, so f_{σ_v} plays the role of α_s. Their centrals sit at the halo center with the halo's velocity: the ELG model has no central velocity bias ($\alpha_c \equiv 0$).

The fitted values (Table 3 and [7]): for LRGs $\alpha_c = 0.33 \to 0.19$ and $\alpha_s = 0.80 \to 0.95$ between the two $z < 0.8$ bins, and for ELGs $f_{\sigma_v} = 1.71^{+0.11}_{-0.14}$ at $\bar{z} = 0.95$ and $1.71^{+0.20}_{-0.15}$ at 1.325 – ELG satellites move *faster* than the dark matter, by 70%. These numbers come only from the

redshift-space fits: the projected clustering w_p integrates the line-of-sight direction away and cannot see velocities.

The ELG occupation. The ELG fits (the mHMQ model [7]: an occupation whose mean central count peaks near a characteristic mass and falls in heavier halos, as expected for star-forming galaxies, with "conformity" letting satellites prefer halos whose central is itself an ELG) give $f_{\rm sat} = 2.6 \pm 0.5\%$ at $\bar{z} = 0.95$ and $3.5 \pm 1.0\%$ at 1.325, with mean host mass $\log_{10}\langle M_h \rangle = 11.78 \to 11.86$ – every change within 1σ. The satellite fraction is model dependent: a phase-space model fitted to DESI Y1 ELGs finds 9.5% [10], nearly four times larger. Section 4.5 shows the project's ELG dispersions sit between the two.

4.3 Step 3: the galaxy-weighted average

The mixture rule. The sample's internal dispersion $\sigma_{\rm HOD}$ averages over two populations with zero-mean velocities: centrals, dispersion $\alpha_c \sigma_{\rm halo}(M_c)$ in their hosts of typical mass M_c, and satellites, $\alpha_s \sigma_{\rm halo}(M_s)$ in heavier hosts M_s. For a mixture of zero-mean populations the *variances* add with the number fractions:

$$\sigma^2_{\rm HOD} \;=\; (1 - f_{\rm sat})\, \alpha_c^2\, \sigma^2_{\rm halo}(M_c) \;+\; f_{\rm sat}\, \alpha_s^2\, \sigma^2_{\rm halo}(M_s). \tag{7}$$

Never average the dispersions themselves. Worked, for the LRG toy at $z = 0.5$ ($f_{\rm sat} = 0.10$, $\alpha_c = 0.3$, $\alpha_s = 0.9$, $M_c = 10^{13.3}$, $M_s = 10^{13.9}\, M_\odot/h$): centrals contribute $0.9 \times (0.3 \times 318)^2 = 0.9 \times 95^2$, satellites $0.1 \times (0.9 \times 506)^2 = 0.1 \times 455^2$, so $\sigma^2_{\rm HOD} = 8{,}190 + 20{,}730 = (170\,{\rm km/s})^2$. The wrong, dispersion-averaged number would be $0.9 \times 95 + 0.1 \times 455 = 131\,{\rm km/s}$, 23% low: the rare, fast satellites dominate the variance (72% of it from 10% of the galaxies), and squaring before averaging is what lets them.

The full mass integral. Eq. (7) compresses an integral over the halo mass function $n(M)$, the number density of halos per unit mass:

$$\sigma^2_{\rm HOD} \;=\; \frac{1}{\bar{n}_g} \int dM\; n(M) \left[\langle N_c \rangle(M)\, \alpha_c^2 + \langle N_s \rangle(M)\, \alpha_s^2 \right] \sigma^2_{\rm halo}(M), \tag{8}$$

with $\bar{n}_g = \int dM\, n(M)\, \langle N_c + N_s \rangle$ the sample's number density. Evaluating it with the Sheth–Tormen mass function [11] built from the note's fiducial $P(k)$ and the $0.4 < z < 0.6$ LRG fit of Table 3 gives $f_{\rm sat} = 0.116$ (the fit's own 0.106 ± 0.012), mean host masses $\log_{10} M = 13.04$ for centrals and 13.82 for satellites, and $\sigma_{\rm HOD} = 171\,{\rm km/s}$ – the two-mass toy's $170\,{\rm km/s}$ to better than one percent. The toy is quantitatively trustworthy because each population's $\sigma^2_{\rm halo}$ is dominated by a narrow range of host masses. The same integral at the $0.6 < z < 0.8$ fit gives $\sigma_{\rm HOD} = 188\,{\rm km/s}$ (these are estimates built on the quoted fits, not published numbers).

Why satellites sit in heavier halos. Figure 3 shows the two host-mass distributions. Centrals pile up just above $M_{\rm cut}$, where halos are still abundant; satellites need $\langle N_s \rangle$ to be appreciable, which happens only towards $M_1 \approx 10\, M_{\rm cut}$, and massive halos host many of them at once. The mean satellite host, $10^{13.8}\, M_\odot/h$, is six times the mean central host – hence nearly twice the virial velocity, and, squared, the dominance in Eq. (7).

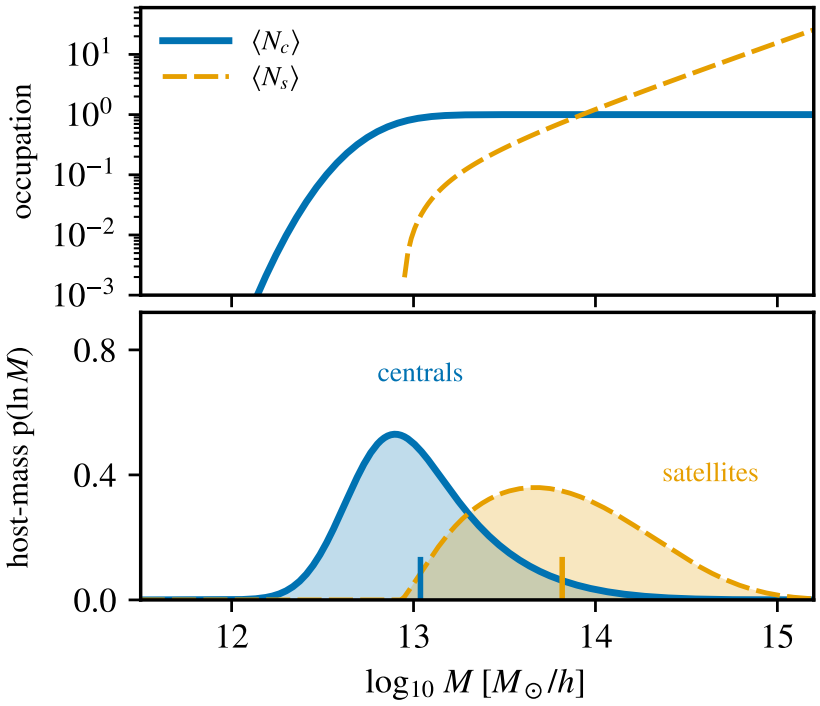


Figure 3: The LRG occupation at $z = 0.5$ (Eq. (6) with the $0.4 < z < 0.6$ fit of Table 3) and the host-mass distributions $p(\ln M) \propto n(M)\, M\, \langle N \rangle$ of its centrals and satellites on a Sheth–Tormen mass function at the note's fiducial cosmology, each normalized to unit area; vertical ticks mark the mean host masses. Satellites occupy halos about six times heavier than centrals.

A subtlety: which average enters the two-halo damping. Eq. (8) weights each galaxy equally. In halo-model redshift space the damping of the *two-halo* (large-scale) power enters differently: each galaxy's Fingers-of-God kernel is carried inside the galaxy-bias integral, weighted by its halo's clustering bias $b(M)$, the mass-dependent version of the sample bias b above, rising with mass [12, 13], and in the low-k expansion the leading damping term is $\propto \int dM\, n(M)\, b(M)\, \langle N_s \rangle\, \sigma_u^2(M)$, with $\sigma_u(M)$ the halo's internal dispersion expressed as a comoving displacement, $(1+z)\,\sigma_{\rm halo}(M)/H(z)$ as in Eq. (1) [14]. Two consequences. Heavier halos, with larger $b(M)$, count more, so the effective two-halo dispersion is somewhat larger than the galaxy-weighted one. And in these models only the satellite term is damped – a central sits at its halo's center and carries no internal displacement – while the measured $\alpha_c > 0$ adds a small central damping on top. The project's blur applies one galaxy-weighted σ_v to the whole bin; the difference between the weightings is a modest, mass-function-weighted systematic that the σ_v priors must cover, not a separate parameter.

4.4 Step 4: the halos' own motion – the linear flow

From the growth of structure to a velocity. On large scales matter conserves mass while it clusters, so the density contrast δ and the peculiar velocity field $\mathbf{v}$ obey the linearized continuity equation $\dot{\delta} = -\nabla\cdot\mathbf{v}/a$, with $a = 1/(1+z)$ the scale factor and the gradient in

comoving coordinates. In linear theory every Fourier mode grows with the same factor $D(z)$ (the linear growth factor, $D(0) = 1$), so $\dot\delta = \delta\,\dot D/D = \delta\,Hf$ with $f \equiv d\ln D/d\ln a$ the growth rate (the chain rule: $\dot D/D = H\,d\ln D/d\ln a$). In Fourier space the divergence becomes $i\mathbf{k}\cdot\mathbf{v}_\mathbf{k}$, and a linear flow is irrotational (any initial rotation decays with the expansion), so $\mathbf{v}_\mathbf{k}$ points along $\mathbf{k}$. Solving the continuity equation for the velocity of mode $\mathbf{k}$,

$$\mathbf{v}_\mathbf{k} \;=\; i\,aHf\,\delta_\mathbf{k}\,\frac{\mathbf{k}}{k^2} \qquad\Longleftrightarrow\qquad \mathbf{v} \;=\; aHf\,\mathbf{\Psi}, \tag{9}$$

where $\mathbf{\Psi}$ is the linear displacement field, the vector whose divergence is $-\delta$. A patch of matter moves with velocity aHf times the displacement that built its environment.

The one-point dispersion. The power spectrum measures the variance of the modes: $\langle\delta_\mathbf{k}\delta^*_{\mathbf{k}'}\rangle = (2\pi)^3\delta_D(\mathbf{k}-\mathbf{k}')\,P_\mathrm{lin}(k)$, with δ_D the Dirac delta. Each mode of Eq. (9) then contributes $(aHf)^2P_\mathrm{lin}(k)/k^2$ to the velocity variance, and summing the modes, $\int d^3k\,(2\pi)^{-3}\,P_\mathrm{lin}(k)/k^2 = (2\pi^2)^{-1}\int P_\mathrm{lin}(k)\,dk$, gives the three-dimensional variance $(aHf)^2(2\pi^2)^{-1}\int P_\mathrm{lin}(k)\,dk$; one line-of-sight component carries a third of it:

$$\sigma_\mathrm{lin}(z) \;=\; \frac{H(z)}{1+z}\,f(z)\,\sigma_d(z), \qquad \sigma_d^2(z) \;=\; \frac{1}{6\pi^2}\int_0^\infty P_\mathrm{lin}(k,z)\,dk \;=\; D^2(z)\,\sigma_{d,0}^2. \tag{10}$$

σ_d is the one-dimensional rms linear displacement. The integral converges at both ends ($P\propto k$ at low k, $\propto k^{n_s-3}\ln^2 k$ at high k) and is dominated by $k\sim 0.01$–$0.3\,h$/Mpc; at the fiducial cosmology $\sigma_{d,0} = 8.86$ Mpc $= 5.96\,\mathrm{Mpc}/h$, stable to 0.03% between upper cutoffs of 10 and 80/Mpc.

Worked values, and why the curve is so flat. At the group redshifts $\bar z_g = 0.18$, 0.33, 0.50, 0.70, 0.95, 1.35 the fiducial gives $\sigma_\mathrm{lin} = 309$, 307, 301, 293, 282, 263 km/s. Three factors nearly cancel: $D(z)$ falls, $f(z)$ rises towards 1 (matter domination), and $H(z)/(1+z)$ passes through a shallow minimum. The net logarithmic slope $d\ln\sigma_\mathrm{lin}/dz$ is -0.01 at $z = 0.18$, -0.12 at 0.5 and -0.17 at 1.35. Cosmology fixes this part, it is smooth in redshift, and the Fisher forecasts hold it at the fiducial (the blur carries no cosmological information by construction) – so *per-bin scatter can live only in the halo part.*

Halos move slightly differently from the matter. σ_lin is the dispersion of a random patch of matter. Halos are not random patches: they form at peaks of the initial field, and a peak's velocity is pulled down by the correlation between its height and the surrounding flow. For Gaussian initial conditions the reduction is the peak-constraint factor $[1-\sigma_0^4/(\sigma_1^2\sigma_{-1}^2)]^{1/2}$ evaluated with the field smoothed on the halo's Lagrangian scale – the comoving radius of the sphere that held the halo's mass before collapse, $R = [3M/(4\pi\bar\rho_m)]^{1/3}$, a few Mpc here [15, 16]. Here $\sigma_j^2 = (2\pi^2)^{-1}\int dk\,k^{2+2j}P(k)W^2(kR)$ are smoothed spectral moments: weighted variances of the field, $j=0$ the density itself, negative j weighing large scales more, with $W(kR)$ the Fourier transform of the smoothing sphere. At the fiducial $P(k)$ the factor is 0.98 at the mean ELG host mass, 0.95 at $10^{13.4}$ and 0.93 at $10^{13.9}\,M_\odot/h$: a few-percent slowdown, growing with mass. Working against it, halo speeds grow *faster* than linear theory in dense regions, by tens of percent for massive halos with massive neighbours [16]. The two corrections pull in opposite directions and neither is computed for the DESI samples, so this note treats the halo-motion part as σ_lin with a several-percent uncertainty; Section 6 charges it to the halo share w as ±0.05.

group	$\bar{z}_g$	σ_v	$\sigma_{\rm lin}(\bar{z}_g)$	$\sigma_{\rm HOD}(\bar{z}_g)$	$w = \sigma^2_{\rm HOD}/\sigma^2_v$
BGS0	0.18	371	309	206	0.31
BGS1	0.33	358	307	185	0.27
LRG1	0.50	357	301	191	0.29
LRG2	0.70	348	293	187	0.29
LRG3	0.95	343	282	196	0.33
ELG1	0.95	304	282	114	0.14
ELG2	1.35	289	263	119	0.17

Table 4: The decomposition of each group's fiducial dispersion at its mean redshift (all velocities in km/s): the linear flow from the fiducial CAMB $P(k)$ (Eq. (10)), the halo part by subtraction ($\sigma^2_{\rm HOD} = \sigma^2_v - \sigma^2_{\rm lin}$), and the halo share w of the variance. $\bar{z}_g$ of the $\Delta z = 0.02$ layout.

4.5 Adding the two parts: $\sigma_v^2 = \sigma_{\rm lin}^2 + \sigma_{\rm HOD}^2$

Why the variances add. The two parts of Eq. (2) are uncorrelated at leading order: whether a satellite is currently moving towards us or away on its orbit does not depend on which way its halo drifts, so $\langle v_{\rm halo}\, v_{\rm int}\rangle = 0$ and

$$\sigma_v^2 \;=\; \sigma_{\rm lin}^2 \;+\; \sigma_{\rm HOD}^2. \tag{11}$$

The argument has limits. Both parts respond to the same environment (halos in dense regions are both faster and more satellite rich), the halo-motion part carries the quasilinear corrections of Section 4.4, and a simulation's measured one-point dispersion contains everything at once. So when this note "inverts" Eq. (11) below, the inferred $\sigma_{\rm HOD}$ is really *everything the linear flow does not explain*, internal motions plus nonlinear flow corrections. With that reading, worked at the toy: $\sqrt{300^2 + 170^2} = 345\,$km/s, squarely inside the LRG fiducials 343–357 km/s. The chain from the virial theorem to the measured dispersions closes.

The decomposition per group. Table 4 inverts Eq. (11) at each group's mean redshift, and Figure 4 does it bin by bin on $\Delta z = 0.02$. The halo share of the variance is $w = 0.27$–0.33 for BGS and LRG and 0.14–0.17 for ELG. These shares control everything in Section 6: a fractional change $\delta\sigma_{\rm HOD}/\sigma_{\rm HOD}$ moves σ_v by only $w\,\delta\sigma_{\rm HOD}/\sigma_{\rm HOD}$, so a 10% halo-part wiggle is a 3% dispersion wiggle for LRGs and 1.5% for ELGs.

Three consistency checks. First, LRG1: the halo model predicts $\sigma_{\rm HOD} = 171\,$km/s (Section 4.3); the subtraction gives 191 km/s – a 10% agreement with no tuning, well inside the velocity-bias error bars and the halo-motion uncertainty. Second, LRG2: the full integral at the $0.6 < z < 0.8$ fit predicts 188 km/s against the subtracted 187 km/s (the exactness is luck; both fits carry $\sim 10\%$ uncertainties). Third, ELG: the subtraction gives 114–119 km/s, while the halo model gives 76 km/s with the One-Percent occupation ($f_{\rm sat} = 0.026$, $f_{\sigma_v}\sigma_{\rm halo}$ at the satellite hosts near $10^{13}\,M_\odot/h$) and 145 km/s with the 9.5% satellite fraction of [10] (both estimates). The project's ELG dispersions sit between the two published occupations: the ELG halo part is real but model dependent at the factor-two level, which is a reason to keep its per-bin freedom fractional in $\sigma_{\rm HOD}$ rather than absolute.

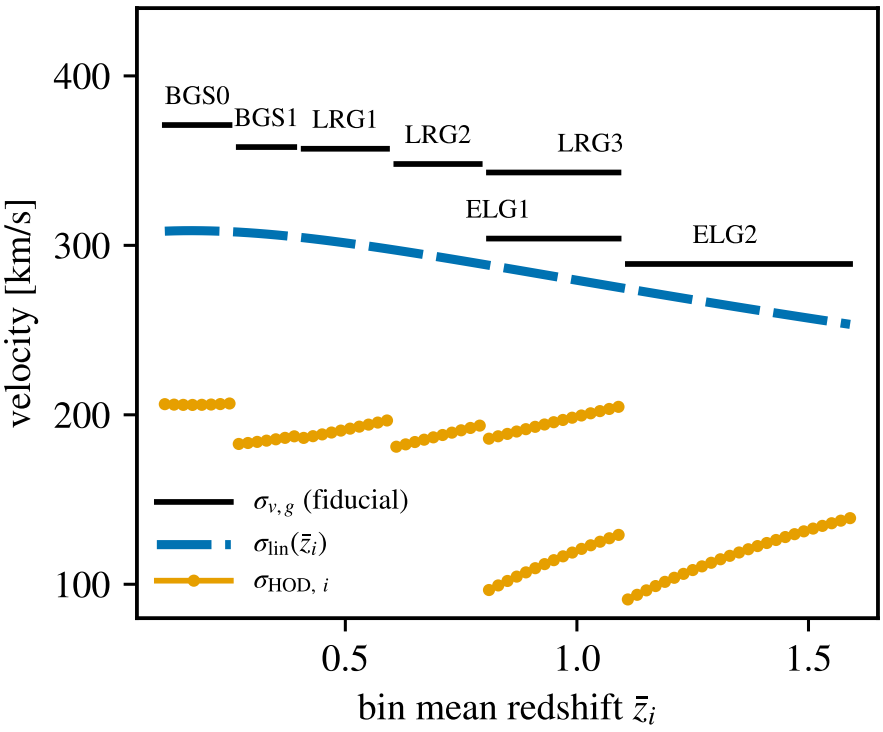


Figure 4: The decomposition per bin on $\Delta z = 0.02$: the fiducial $\sigma_{v,g}$ (flat per group), the linear flow $\sigma_{\rm lin}(\bar{z}_i)$, and the implied halo part $\sigma_{{\rm HOD},i} = (\sigma^2_{v,g} - \sigma^2_{\rm lin}(\bar{z}_i))^{1/2}$. Where two groups share a redshift range (LRG3 and ELG1), the ELG group is the lower pair of lines. The rise of $\sigma_{{\rm HOD},i}$ inside each group is the $T_{\rm lin}$ truth of Section 7: a flat σ_v forces the halo part to climb against the falling linear flow.

What the halo model then predicts for redshift trends. Between the two LRG ξ fits the full integral's $\sigma_{\rm HOD}$ rises from 171 to 188 km/s over $\Delta z = 0.2$: a slope $d\ln\sigma_{\rm HOD}/dz \approx +0.5$ per unit z (estimate). The two-mass toy decomposes the competing effects: the host-mass decline ($\log_{10}\overline{M}_h$ 13.42 $\to$ 13.00 across $\bar{z} = 0.5 \to 1.03$) with $\sigma_{\rm halo} \propto (ME(z))^{1/3}$ gives $d\ln\sigma_{\rm halo}/dz \approx -0.41$, while the rising satellite fraction (8.9 $\to$ 15.1%, w_p series) gives about $+0.4$, and the fitted velocity-bias changes (α_s up, α_c down) partly cancel each other (+13% and −11% on $\sigma_{\rm HOD}$). The honest conclusion: the sign of the LRG slope is not established, and magnitudes up to $|d\ln\sigma_{\rm HOD}/dz| \sim 0.5$ are easy to produce. That is the slope prior's scale.

4.6 One-point versus pairwise dispersions

Two different statistics. The blur needs the *one-point* dispersion: the rms velocity of single galaxies. Most measurements quote a *pairwise* dispersion $\sigma_{12}(r)$: the rms of the velocity difference between two galaxies separated by r. For zero-mean line-of-sight velocities with one-point variance $\sigma^2_{\rm 1pt}$ and two-point correlation $\psi(r) = \langle v_1 v_2\rangle$,

$$\sigma^2_{12}(r) \;=\; \big\langle (v_1 - v_2)^2 \big\rangle \;=\; 2\,\big[\sigma^2_{\rm 1pt} - \psi(r)\big]. \tag{12}$$

At small separations the coherent flow moves both members of a pair together, so its contribution cancels ($\psi \to$ the flow's variance) and what survives is twice the *internal* variance of the pair's halo; at large separations $\psi \to 0$ and $\sigma_{12} \to \sqrt{2}\,\sigma_{1\mathrm{pt}}$. On top of that, pairs are counted per pair, $\propto N(N-1)$, so a halo with N satellites contributes $N(N-1)/2$ one-halo pairs: rich clusters dominate pairwise statistics far more than they dominate the galaxy count.

What the classic numbers actually are. In CfA2+SSRS2, $\sigma_{12} = 540 \pm 180\,\mathrm{km/s}$ at $r \sim 1\,\mathrm{Mpc}/h$ [17]; removing the Abell $R \geq 1$ clusters drops it to $295 \pm 99\,\mathrm{km/s}$, and the subvolume-to-subvolume scatter is dominated by whether a rich cluster fell in the volume. In GAMA the small-scale pairwise dispersion rises from about $200\,\mathrm{km/s}$ at $M_r = -17$ to about $600\,\mathrm{km/s}$ at $M_r \approx -22$ [18], while HOD models predict the opposite luminosity trend at small and large r, driven by the falling satellite fraction above L_* [19]. The BOSS full-shape fits give damping parameters $\sigma_v = 6.10 \pm 0.69$, $5.84^{+0.70}_{-0.77}$ and $5.35^{+0.76}_{-0.81}\,\mathrm{Mpc}/h$ (NGC) at $z_{\mathrm{eff}} = 0.38$, 0.51, 0.61 [20], and VIPERS gives $\sigma_{12} = 5.00 \pm 0.86$ and $3.54 \pm 0.78\,\mathrm{Mpc}/h$ at $z = 0.6$ and 0.86 [21]; converted with Eq. (1) at the note's fiducial these are 539, 509, 464 km/s and 434, 309 km/s (estimates). All of these are *model* parameters of pairwise-velocity damping templates, fitted alongside bias and shot noise (the Poisson noise of counting a finite number of galaxies), not direct one-point measurements.

Why none of them is σ_v. Compare 540 km/s (CfA2, pair weighted, cluster dominated) with the project's 371–289 km/s (one point, galaxy weighted): the factor $\sqrt{2}$ of Eq. (12), the $N(N-1)$ weighting towards rich halos, and the model dependence each move the number by tens of percent. The literature can therefore *bracket* redshift trends and scatter (Section 5) but cannot replace the source forecast's own one-point measurement – which is why the priors of Section 6 stay anchored to it.

4.7 Caveats: the blur's shape, the Kaiser boundary, and SN_2

A mixture is not a Gaussian. The blur convolves $n_i(z)$ with one Gaussian of width $\sigma_{z,i}$. But Section 4.3 built the velocity distribution as a mass-weighted mixture of Gaussians of very different widths (95 km/s centrals, 455 km/s satellites), and a mixture has heavier tails than one Gaussian of the same total variance – summed over the halo mass function the tails are close to exponential [16]. The project blurs with one Gaussian at the mixture's total variance. How much the tail shape matters for band powers is an open robustness check (the Fingers-of-God profile item in the appendix), not settled here.

Possible double counting with the Kaiser term. The one-point σ_v contains the full linear flow, $\sigma^2_{\mathrm{lin}} = \sigma^2_v - \sigma^2_{\mathrm{HOD}}$ by construction. The coherent part of that same flow is what the Kaiser term models as a systematic distortion of the clustering amplitude. Blurring the window with the *full* σ_v treats the whole velocity as random scatter uncorrelated with the density, so the correlated large-scale part enters twice: once coherently (Kaiser), once as damping (the blur). Eq. (12) shows the size of the ambiguity: a damping built from pair separations would subtract $\psi(r)$, which at small r removes most of σ^2_{lin} – precisely the difference between damping with σ^2_v and with σ^2_{HOD}. The project follows the source forecast and uses the full measured dispersion; the sixfold prior widths, introduced for the α_2 degeneracy, also cover this modeling

ambiguity. This note states the issue and does not settle it; the study's design (per-bin scatter on the halo part only) is unaffected, because both conventions share the halo part.

The SN_2 stochastic term carries the same satellite velocities. The project's one-loop bias model (its perturbation-theory model of the galaxy power spectrum, `notes/higher_order_bias`) carries a line-of-sight stochastic term $\mathrm{SN}_2\, k^2\mu^2$, with μ the cosine of the angle between $\mathbf{k}$ and the line of sight, in units of $f_{\rm sat}\sigma^2_{\rm sat}/\bar{n}_i$, the DESI DR1 scaling of Maus et al. [14]: in their halo-model expansion the one-halo (central–satellite and satellite–satellite) power at low k contributes a $k^2\mu^2$ white term (white: a constant amplitude in k, like shot noise) of amplitude $f_{\rm sat}\,\sigma^2_{1,\rm eff}/\bar{n}_g$, with $\sigma_{1,\rm eff}$ the $\langle N_c N_s\rangle$-weighted satellite dispersion in distance units. The code's unit is their stochastic-prior table: LRG $(f_{\rm sat}, \sigma) = (0.1, 7.8\,\mathrm{Mpc}/h)$, ELG $(0.1, 2.9\,\mathrm{Mpc}/h)$, BGS borrowing the LRG row (`SN2_UNIT` in `likelihood/desi_cl.py`; 6.084 and 0.841 $(\mathrm{Mpc}/h)^2$). Converted with Eq. (1), $7.8\,\mathrm{Mpc}/h$ is 677–684 km/s across the LRG groups – about 1.5 times the halo-model satellite dispersion $\alpha_s\sigma_{\rm halo}(M_s) \approx 430$–$455\,\mathrm{km/s} \approx 5\,\mathrm{Mpc}/h$ (Section 4.3; estimates), so Maus et al.'s number is a generous order-of-magnitude width, not a measurement. The ELG unit assumes $f_{\rm sat} = 0.1$, four times the One-Percent fit. The physical point: the blur (window side) and SN_2 (one-halo stochastic side) describe the same satellite virial motions in two places the projection treats differently; like the α_2 degeneracy of the Fingers-of-God note, this is a family of partially redundant descriptions, and the priors, not a fit, set how the redundancy is shared.

5 What the literature says about slopes, scatter and correlation lengths

No published analysis measures the Fingers-of-God dispersion in thin redshift slices for any DESI tracer. What exists is HOD fits in broad bins ($\Delta z = 0.2$–0.5), damping fits in broad bins, and this note's own propagation of them. The evidence for each number below is graded explicitly.

Slopes within a sample. The linear part falls at $d\ln\sigma_{\rm lin}/dz = -0.01$ to -0.17 per unit z (fiducial $P(k)$, firm). For the halo part: the LRG HOD changes between broad bins give slopes of either sign up to $|d\ln\sigma_{\rm HOD}/dz| \approx 0.5$ (Section 4.5; estimate). The BOSS damping values above fall as $d\ln\sigma/dz \approx -1.0 \pm 0.6$ per unit z (a straight-line fit to the NGC and SGC values; estimate, 1–1.5σ, model-dependent damping of a mixed LOWZ+CMASS sample), and VIPERS gives -1.3 ± 1.0 (same caveats). In σ_v units every halo-part slope is multiplied by w. For a flux-limited BGS (a sample limited by apparent brightness, so at higher redshift only intrinsically brighter galaxies make the cut and the population itself drifts) the *sign* is open: the luminosity limit brightens by about 2.5 mag across $0.1 < z < 0.4$ (estimate), GAMA's pairwise dispersion rises with luminosity while HOD models predict it falls [18, 19], and the project already treats BGS as flux limited (its bias drift in `scripts/make_desi_params.py` comes from the flux limit).

Where departures cluster: selection edges. The sharpest documented population changes sit where the target selection changes, within $\Delta z \approx 0.1$–0.15: the DESI LRG completeness collapse at $z > 0.95$ (Table 3); and BOSS in thin shells, where no smooth $b(z)$

model fits above $z \approx 0.6$, the onset of flux-limited behavior, forcing that analysis to drop its last three shells [22]. So population departures correlate over $\Delta z \sim 0.1$–0.3 inside a group and over 0.05–0.1 near sample edges. No paper measures a correlation length for a dispersion; L is argued from the redshift scale of selection changes, not measured.

A white (uncorrelated) part. A finite slice holds a finite number of rich halos, and Section 4.6 showed how strongly they weight velocity statistics: in CfA2+SSRS2 the spread between subvolumes of about $7 \times 10^5\,(\mathrm{Mpc}/h)^3$ was cluster dominated [17]. Scaling that spread as $V^{-1/2}$ to a 62 Mpc slice of the DESI footprint (about $10^8\,(\mathrm{Mpc}/h)^3$) leaves $\lesssim 3\%$ on the pair-weighted statistic, and less on the galaxy-weighted σ_v, which rich halos dominate more weakly: $\lesssim$ 1–2% of σ_{HOD} per bin, with L of about one bin (crude estimate).

Thin-bin precedents. The BOSS tomographic analysis closest in spirit to this project [22] used 18 thin shells with *one* Fingers-of-God parameter a_{vir} scaled deterministically as $(D(z)/D(z_{\mathrm{ref}}))^2$ and one linear bias law $b_1 + b'(z - z_{\mathrm{ref}})$: no parameter was free per shell. For the harmonic-space power spectrum, the Fingers of God suppress C_ℓ on *all* transverse scales once the bin is thinner than $\Delta r \lesssim \pi\sigma_u$, with σ_u the dispersion in distance units [23] – for $\sigma_v = 350\,\mathrm{km/s}$ at $z = 0.5$, $\pi\sigma_u \approx 19$ Mpc, below the 62 Mpc bins but not by orders of magnitude; and for $\Delta z \lesssim 0.01$ nonlinear redshift-space corrections enter at all ℓ [24]. No published exponential-kernel or Gaussian-process prior on a Fingers-of-God parameter across thin bins was found: the construction below appears to be new in this context.

6 The per-bin model

Where the freedom lives. Write each bin's dispersion as Eq. (11) with the linear part held at the fiducial cosmology. The fiducial stays exactly today's: $\sigma_{v,i} = \sigma_{v,g}$, flat in each group, so every existing derivative base remains valid; the implied fiducial halo part $\sigma^2_{\mathrm{HOD},i} = \sigma^2_{v,g} - \sigma^2_{\mathrm{lin}}(\bar{z}_i)$ rises slightly across each group (Figure 4). The group *level* keeps today's parameter and prior (the paper widths of Table 2). The new freedom acts on the halo part only:

$$\sigma_{\mathrm{HOD},i} \;\longrightarrow\; \sigma_{\mathrm{HOD},i}\,\big[\,1 + s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i\,\big], \tag{13}$$

with s_g a free slope per group and ε_i a correlated per-bin residual.

The residual prior, borrowed from the linear bias. ε has the exponential-kernel prior $\langle \varepsilon_i \varepsilon_j \rangle = \sigma^2 e^{-|\bar{z}_i - \bar{z}_j|/L}$ within a group (zero across groups; L a correlation length in redshift: residuals of bins closer than L move together), *projected* so that it charges (penalizes in the χ^2) neither the group's level nor its slope: the prior precision is $P = Q - QU(U^{\mathsf{T}}QU)^{-1}U^{\mathsf{T}}Q$ with Q the inverse kernel and $U = [1,\, z - \bar{z}_g]$, the construction built and derived step by step for the linear bias b_1 in the correlated-prior note, `notes/correlated_bias_prior_v2` (its appendix sections on the projected precision and the memory projection). The level and the slope stay free (the level under today's Gaussian prior); the kernel constrains only the wiggles around each group's line.

piece	prior	where the number comes from
level	today's Gaussian, paper widths	Table 2
slope s_g	$N(0, 0.5/1.0/2.0)$ per unit z; BGS 2.0	Section 4.5; BGS sign open
residual A	$\sigma = 10\%$ of $\sigma_{\rm HOD}$, $L = 0.2$	broad-bin HOD changes (7–21%)
residual B	$\sigma = 20\%$ (30% BGS), $L = 0.05$	selection edges (LRG $z >$ 0.95; BOSS $z > 0.6$)
white term (optional)	5% of $\sigma_{\rm HOD}$, $L = 0.02$	rich-halo shot noise (crude)

Table 5: The per-bin priors on the halo part and their provenance. In σ_v units each width is multiplied by the halo share ($w \approx 0.27$–0.33 BGS/LRG, 0.14–0.17 ELG): residual A is a 3% dispersion wiggle for LRGs and 1.4–1.7% for ELGs.

The linear response, and why no new physics is needed. Differentiating $\sigma_{v,i}^2 = \sigma_{{\rm lin},i}^2 + \sigma_{{\rm HOD},i}^2$ at fixed $\sigma_{\rm lin}$, $\delta\sigma_{v,i} = (\sigma_{{\rm HOD},i}^2/\sigma_{v,i})\,(\delta\sigma_{{\rm HOD},i}/\sigma_{{\rm HOD},i})$, so a fractional halo-part change reaches the sampled dispersion suppressed by the halo share:

$$\delta\sigma_{v,i} \;=\; w_i\,\sigma_{v,g}\,\big[\,s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i\,\big], \qquad w_i = \frac{\sigma_{{\rm HOD},i}^2}{\sigma_{v,g}^2}. \tag{14}$$

Every variant of the study is therefore a linear combination of per-bin σ_v derivative columns (a derivative column is the derivative of the whole data vector with respect to one parameter, the building block of the Fisher matrix; the machinery note builds everything from them) – the slope and residual directions are those columns weighted by $w_i\sigma_{v,g}$ – and the likelihood needs no new option: the freedom costs only the per-bin columns (Section 7). Worked sizes from Table 4: a one-sigma baseline slope ($s_g = 1$ per unit z) tilts LRG3's dispersions by $w\,\Delta z \approx 0.33 \times (\pm 0.14) \approx \pm 4.6\%$ of $\sigma_{v,g}$, about ± 16 km/s across the group; the same slope moves ELG1 by only $\pm 1.9\%$, the halo share being half as large.

How each prior number follows from the physics. Table 5 summarizes; the trail:

- **Slope widths** 0.5/1.0/2.0**.** Section 4.5 produced halo-part slopes of either sign up to $|d\ln\sigma_{\rm HOD}/dz| \approx 0.5$ from fitted HOD evolution alone; 0.5 is therefore optimistic (the halo model taken at face value), 1.0 baseline (covering the BOSS and VIPERS damping trends at about 1σ once multiplied by w), 2.0 pessimistic. BGS takes 2.0 because its flux-limited slope has an open sign (Section 5).

- **Residual A (**10%**,** $L = 0.2$**): population drift.** The HOD changes the DESI fits find between broad bins move $\sigma_{\rm HOD}$ by +7% ($f_{\rm sat}$ 9% → 11%), +13% (α_s 0.80 → 0.95), −11% (α_c 0.33 → 0.19) and +21% ($f_{\rm sat}$ 8.9 → 15.1%) in the two-mass toy; 10% with $L = 0.2$ lets such drifts happen on the $\Delta z \sim 0.2$–0.4 scale they are observed on.

- **Residual B (**20%**, BGS** 30%**,** $L = 0.05$**): selection edges.** Population changes concentrate where the selection changes within $\Delta z \approx 0.15$ (Section 5); 20% of $\sigma_{\rm HOD}$ is the size of the LRG $z > 0.95$ upheaval, and 30% for BGS reflects the open sign and the stronger selection drift of a flux-limited sample.

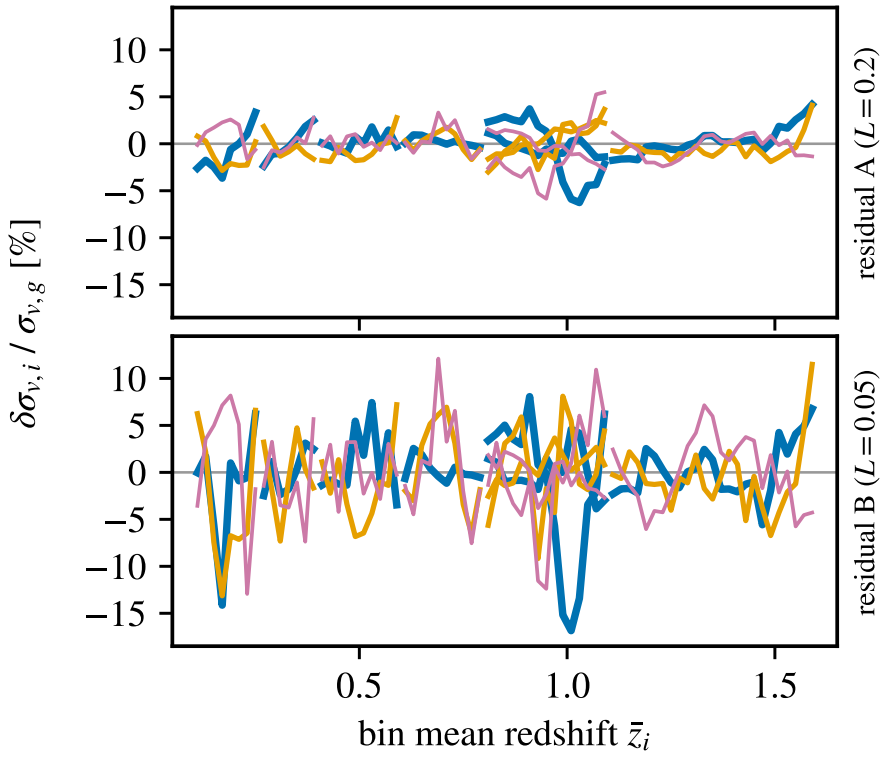


Figure 5: Random truths of the per-bin model on $\Delta z = 0.02$, as percent of each group's $\sigma_{v,g}$ (Eq. (14); baseline slope width, kernel-projected residuals): three draws under residual A (top) and residual B (bottom), the same slope draws in both panels, so only the residual changes between them. The largest excursions sit where the halo share w is largest (BGS and LRG3); the ELG groups, with $w \approx 0.15$, barely move under A.

- **White term (5%, $L = 0.02$).** The rich-halo shot noise estimate of Section 5, roughly one bin in correlation length. It is a crude number and the term is optional.

What the draws look like. Figure 5 shows random truths under slope + A and slope + B. A produces slow, few-percent breathing of a group's dispersions; B produces bin-to-bin structure reaching ±10% of σ_v in the high-w groups. For comparison, the per-bin error bars of the source forecast's measurement are 2–4% of σ_v for LRG and ELG and 6–8% for BGS: residual A is at the measurement's own resolution, residual B deliberately above it.

Robustness of w. The halo shares inherit the halo-motion uncertainty of Section 4.4 (peaks slower by 2–7%, nonlinear evolution faster by a similar amount, with opposite signs), so w per group is uncertain by roughly ±0.05. Since Eq. (14) uses w_i only as a fixed weight on the derivative columns, a change of w touches no evaluation, only the weights; the appendix records why no separate refit is needed.

7 The study

The accuracy test comes before any FoM. A figure of merit from a model that cannot absorb the true signal is meaningless, so the project always tests accuracy first (the linear-bias ladder of `notes/correlated_bias_prior_v2` set the pattern). Fit today's model V0, with its flat per-group σ_v, to data whose true dispersions vary bin by bin, compute each scored parameter's shift through the linear response, and measure how well each fit's quoted error bar covers where the fit actually lands: $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, with $\chi^2 = \Delta p^{\mathsf{T}} C_p^{-1} \Delta p$ (recovered minus true, the fit's marginalized covariance – marginalized: the errors with every other parameter left free; $n_{\rm dof} = 2$ for (w_0, w_a), 1 for Ω_m or A_s; $\eta = 1$ means honest error bars).

The truths.

- $T_{\rm lin}$**: the linear flow's own redshift dependence.** This truth needs no prior choice at all. The fiducial holds σ_v flat in each group, but Section 4.4 fixed the linear part's decline; a truth with a flat halo part instead, $\sigma^2_{v,i} = \sigma^2_{\rm lin}(\bar z_i) + [\sigma^2_{v,g} - \sigma^2_{\rm lin}(\bar z_g)]$, differs from the fiducial by $\delta\sigma_{v,i} \simeq [\sigma^2_{\rm lin}(\bar z_i) - \sigma^2_{\rm lin}(\bar z_g)]/(2\sigma_{v,g})$ (linearized: $\sqrt{\sigma^2_{v,g} + x} \simeq \sigma_{v,g} + x/(2\sigma_{v,g})$ for $|x| \ll \sigma^2_{v,g}$). At the fiducial $P(k)$ this is at most ± 3 km/s for LRG1–2, ± 5–6 km/s for LRG3 and ELG1, and -9 to $+10$ km/s across ELG2's 25 bins – up to 3.5% of $\sigma_{v,g}$, purely from cosmology. If V0 is biased even under $T_{\rm lin}$, per-bin freedom is required, whatever the priors.
- **Physical draws.** Random truths from the slope prior plus residual A, and from the slope prior plus residual B (with and without the white term), as in Figure 5, scored as an ensemble.

The variants. With the accuracy test passed or failed on record, the cost of the freedom is measured on the four layouts and the grid of scale cuts ($k_{\rm max}$ quoted at $z = 0$; each bin keeps angular scales down to its own cut $k_{\rm max}/D(\bar z_i)$, the machinery note's criterion), for FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$:

V0	today's model: one σ_v per group
V1	+ the halo-part slope s_g (three widths)
V2	+ residual A
V3	+ residual B (and B + white)
V4	every per-bin σ_v free apart from the level prior (the bound)

V4 is the bound: the most freedom the family allows, hence the lowest FoM. It needs no w: once each bin is free, the halo share no longer enters. A variant that fails the accuracy test gets no FoM in any figure.

Implementation, in brief. (1) $\sigma_{\rm lin}(\bar z_i)$ and w_i were computed for every bin of the four layouts from the fiducial $P(k)$ (seconds; this note's generators already did it for $\Delta z = 0.02$), and $w_i > 0$ holds everywhere. (2) The per-bin step machinery of `scripts/compute_fisher_base_mpi_classes.py` was extended with a `"sigv"` operator: a relative step on the per-bin `DESI_SIGV_<n>` of one color of the greedy coloring (the script's batching of bins into sets stepped in one evaluation, kept in bin order), with the same five-point finite-difference stencils as the

b_1 columns (the machinery note derives the rule). (3) The columns were computed at three relative steps, 0.01, 0.02 and 0.04, with 0.02 the main step; the step test is in Section 8.4. (4) The new explicit caches `model_sigv_bins_<layout>_final` hold the per-bin columns at all three steps; today's caches were not touched. (5) The sum of a group's per-bin columns reproduces today's group `DESI_SIGV` column (Section 8.4). (6) The accuracy test ran first, then the variants. Each layout cost 50 model evaluations – one at the fiducial, four colors times four stencil points times three steps, and one at the fiducial again as a reset check – and, on AC power, 9 minutes of wall time on $\Delta z = 0.2$, 17 on $\Delta z = 0.05$, 36 on $\Delta z = 0.02$ and 35 on $\Delta d_{\rm com} = 62$ Mpc (one MPI rank with eight threads, because of memory); the fits take seconds. The fits live in `scripts/compute_sigv_bins_fits.py`, whose result files the generator `plots/make_sigv_results_data.py` copied into the figure folders of Section 8.

8 Results

What there is to vary, layout by layout. All seven dispersion groups exist on every layout, but the bins they hold differ enormously. On $\Delta z = 0.2$ most groups hold a single bin (LRG3, ELG1 and ELG2 hold two), so a per-bin model has almost nothing to vary there: every shift, η and FoM loss of this section is 0 or 1 on that layout to better than 10^{-4}, and the figures omit it where that is so (each caption says what is omitted). On $\Delta z = 0.02$ the $k_{\rm max} = 0.05\,h$/Mpc cut keeps no band at all, so that layout's curves start at 0.075. The $T_{\rm lin}$ truth itself is small in absolute terms: the largest per-bin difference between it and the flat fiducial is 9.9 km/s on $\Delta z = 0.02$ (3.4% of $\sigma_{v,g}$), 8.8 km/s (3.0%) on $\Delta d_{\rm com} = 62$ Mpc, 9.2 km/s (3.2%) on $\Delta z = 0.05$ and 4.1 km/s on $\Delta z = 0.2$.

8.1 The accuracy test under $T_{\rm lin}$

A deterministic shift, not a scatter. $T_{\rm lin}$ involves no prior and no random draw: the linear flow's dispersion declines with redshift across each group (Section 4.4) while the model holds σ_v flat, so the mismatch between truth and model is one definite vector whose sign and size are fixed by the fiducial cosmology. Fitting V0 to this truth therefore displaces the best fit by a definite amount – a shift, not a widening of the scatter – and Figure 6 reports that shift against the scale cut. Because the paper is a Fisher forecast, these shifts are the estimate of what an MCMC run on such data would see, not a pass/fail test.

The numbers. On $\Delta z = 0.02$ the (w_0, w_a) fit moves by 0.06 of its error ellipse at $k_{\rm max} = 0.075$, 0.18 at 0.1 and 0.32 at 0.2; A_s moves by 0.04–0.14 σ and Ω_m by 0.01–0.06 σ. On $\Delta d_{\rm com} = 62$ Mpc the ellipse shift is 0.007 at 0.05, 0.05–0.06 at 0.075–0.125 and about 0.10 at 0.15–0.2, with A_s up to 0.07 σ and Ω_m up to 0.03 σ. On $\Delta z = 0.05$ the ellipse shift stays at 0.001–0.010 and the single parameters below 0.007 σ; on $\Delta z = 0.2$ everything is below 2×10^{-4}.

Why the thin layouts are the sensitive ones. The ordering of the layouts is the edge-fraction argument of the cross-correlation note: a Gaussian blur of width σ_z moves a fraction of about $\sigma_z/(\sqrt{2\pi}\, w_{\rm bin})$ of a bin's galaxies across each edge, with $w_{\rm bin}$ the bin's width in redshift, so the thinner the bin, the larger the share of its galaxies the blur relocates – and the more strongly its spectra respond to its σ_v. The widths, worked at the fiducial cosmology: a Δz bin is $c\,\Delta z/H(z)$ deep, and with $H(0.95) = 115$ km/s/Mpc a $\Delta z = 0.02$ bin is $299792 \times 0.02/115 \approx$

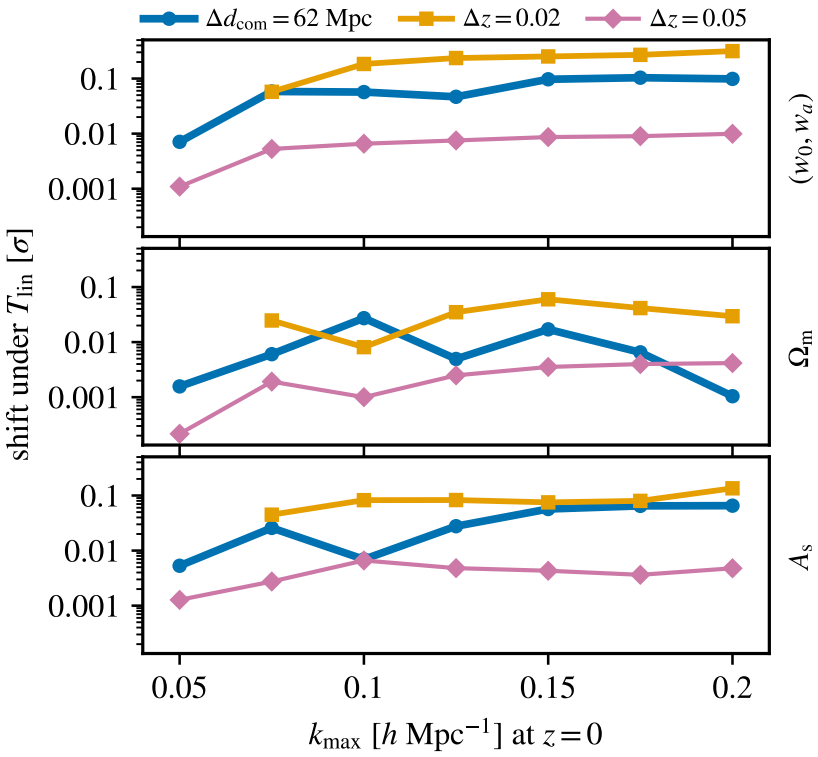


Figure 6: The shift of today's model V0 when the true dispersions follow the linear flow's own redshift dependence (the truth $T_{\rm lin}$), against the scale cut. Top to bottom: (w_0, w_a) in units of its error ellipse, $\sqrt{\Delta p^{\mathsf{T}} C^{-1} \Delta p}$ with C the pair's 2×2 marginalized covariance, then Ω_m and A_s in units of their own σ. The $\Delta z = 0.2$ layout is not drawn: its shifts stay below $10^{-3}\,\sigma$. On $\Delta z = 0.02$ no band survives the $k_{\rm max} = 0.05\,h$/Mpc cut, so its curve starts at 0.075. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

52 Mpc deep at $z = 0.95$, falling to 37 Mpc at $z = 1.55$ ($H = 160\,$km/s/Mpc): thinner than the fixed 62 Mpc bins exactly where the ELG groups put most of their bins, which is why $\Delta z = 0.02$ overtakes $\Delta d_{\rm com} = 62$ Mpc in Figure 6. The same arithmetic makes a $\Delta z = 0.05$ bin 90–130 Mpc deep there, twice the 62 Mpc width, and a $\Delta z = 0.2$ bin several hundred Mpc deep. The shift also grows with the cut, because raising $k_{\rm max}$ admits more bands, and the bands it adds are the small-scale ones where the blur's suppression – the Fingers-of-God damping of Section 3 – is strongest, so the data's sensitivity to each bin's σ_v grows with it.

8.2 Error bars under the random truths

The truth and the baseline. The ensemble truth $T_{\rm lin} + \mathrm{B}$ adds to the fixed $T_{\rm lin}$ shape random draws of the halo-part slope (at the baseline width 1.0) and of residual B, as in Figure 5. As in the project's other notes, the ensemble also draws the linear-bias truths of the paper prior and the data noise, so the test scores the whole fit, not the dispersions in isolation; with no σ_v truth at all it returns $\eta = 1.0000$ on every layout and cut – the honest baseline every departure below is measured from. Figure 7 shows η for V0 and for V1 at the baseline

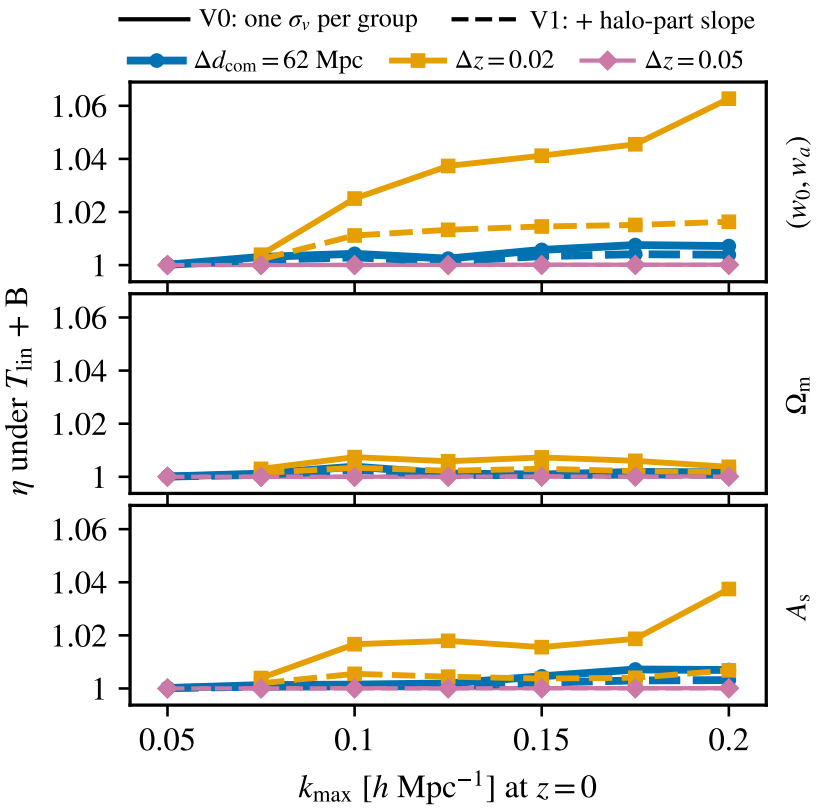


Figure 7: How well each fit's quoted error bar covers where the fit actually lands, measured by η ($\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ with $\chi^2 = \Delta p^{\mathsf{T}} C_p^{-1} \Delta p$, Section 7; $\eta = 1$ means honest error bars), under the truth $T_{\rm lin}$ + B, against the scale cut. Today's model V0 (solid) and the slope variant V1 at the baseline width (dashed); top to bottom: (w_0, w_a), Ω_m, A_s. The $\Delta z = 0.2$ layout is not drawn: its η differs from one by less than 10^{-5}. The other slope widths and residual variants are quoted in the text. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

slope width.

The numbers. For V0 on $\Delta z = 0.02$, η of (w_0, w_a) is 1.004 at $k_{\rm max} = 0.075$, 1.025 at 0.1, 1.037–1.046 at 0.125–0.175 and 1.063 at 0.2; A_s reaches 1.037 and Ω_m 1.007. On $\Delta d_{\rm com} = 62$ Mpc V0 stays at or below 1.0076 for (w_0, w_a), 1.007 for A_s and 1.004 for Ω_m; on $\Delta z = 0.05$ at or below 1.0002. The slope alone repairs most of this: V1 at the baseline width keeps (w_0, w_a) at 1.002–1.016 on $\Delta z = 0.02$ (A_s at most 1.007, Ω_m at most 1.003) and at most 1.004 on $\Delta d_{\rm com} = 62$ Mpc. At the hardest point – $k_{\rm max} = 0.2$, $\Delta z = 0.02$, (w_0, w_a) – the other variants: V1 at width 0.5 gives 1.043, at width 2.0 0.998 (error bars slightly wider than the actual scatter); V2 (slope + residual A) 1.016; V3 (slope + residual B) 1.014, unchanged by the white term; V4 1.0001. Under the random draws alone, with no $T_{\rm lin}$, the same point gives V0 1.039, V1 1.005 and V3 1.004; under $T_{\rm lin}$ alone V0 gives 1.025. Both loads matter for today's model. The quantity that adds over independent truths is $\eta^2 - 1$ (the truths' contributions to $\langle\chi^2\rangle$ add), and for V0 the random draws contribute $1.039^2 - 1 \approx 0.080$ and

$T_{\rm lin}$ $1.025^2 - 1 \approx 0.050$, which together give $\sqrt{1.130} \approx 1.063$. The two loads respond differently to the slope: V1 removes almost all of the random load, while most of the $T_{\rm lin}$ shift survives it. Isolating that shape, the $T_{\rm lin}$ shift of (w_0, w_a) at this point falls along the family as 0.32 (V0), 0.28 (V1, width 0.5), 0.21 (width 1.0), 0.12 (width 2.0), 0.21 (V2 and V3) and 0.011 (V4).

Why the slope absorbs only part of $T_{\rm lin}$. In halo-part units the $T_{\rm lin}$ tilt is large, because the halo share w is small. Worked on ELG2 with Table 4 and the group ranges of Section 2: ELG2 has $w = 0.17$ and 25 bins spanning $\Delta z \approx 0.5$, and $T_{\rm lin}$ moves its dispersions by up to 3.4% of $\sigma_{v,g}$ at the group ends (the -9 to $+10\,$km/s of Section 7). Dividing by w, that is a $3.4/0.17 \approx 20\%$ tilt of $\sigma_{\rm HOD}$ across half the group (0.25 in z): a slope of about 0.8 per unit z, close to one sigma of the baseline slope prior. A Gaussian prior pulls the fitted slope back toward its center at zero, so V1 absorbs only part of the tilt; the 2.0 width absorbs more, and V4, with no shape prior at all, absorbs essentially all of it. The residuals cannot absorb a tilt either: their directions are projected so that they carry neither a level nor a slope component (Section 6), which is why V2 and V3 sit on top of V1 under $T_{\rm lin}$.

8.3 The cost of the freedom

The numbers. Figure 8 gives the price, as the loss of each FoM against V0. V1 at the baseline width costs, on $\Delta z = 0.02$, 0.3% of FoM(w_0, w_a) at $k_{\rm max} = 0.075$, 1.6% at 0.1, 2.3–2.6% at 0.125–0.175 and 3.4% at 0.2 (2.6% of FoM$(\Omega_m, 10^9 A_s)$ there); on $\Delta d_{\rm com} = 62$ Mpc at most 0.5%, on $\Delta z = 0.05$ at most 0.011% and on $\Delta z = 0.2$ below 3×10^{-5} percent. The other variants at $k_{\rm max} = 0.2$, FoM(w_0, w_a): on $\Delta z = 0.02$, V1 at width 0.5 costs 1.2%, at width 2.0 6.2%, V2 3.4%, V3 3.6%, V3 with the white term 3.6%; on $\Delta d_{\rm com} = 62$ Mpc, V1 at width 2.0 costs 1.3% and V3 0.5%; on $\Delta z = 0.05$ everything stays at or below 0.04%. V4, the bound, costs at $k_{\rm max} = 0.2$: 6.1, 11.2, 5.0 and 0.4% of FoM(w_0, w_a) and 6.4, 8.5, 7.7 and 2.9% of FoM$(\Omega_m, 10^9 A_s)$ on $\Delta d_{\rm com} = 62$ Mpc / $\Delta z = 0.02$ / 0.05 / 0.2. At the low cuts the bound is expensive: 59% ($\Delta d_{\rm com} = 62$ Mpc at 0.05), 48% ($\Delta z = 0.02$ at 0.075) and 30% ($\Delta z = 0.05$ at 0.05) of FoM(w_0, w_a), and up to 72% of FoM$(\Omega_m, 10^9 A_s)$ ($\Delta d_{\rm com} = 62$ Mpc at 0.05).

Why the residuals are nearly free, and the bound is not. V2 and V3 cost at most 0.2 percentage points more than V1: their prior is projected (no level, no slope) and tight once expressed in σ_v units – a residual of 20% of $\sigma_{\rm HOD}$ reaches σ_v multiplied by the halo share, and $w \times 20\%$ is a few percent (Table 5) – so the wiggle directions they open are directions the prior immediately closes. V4 is expensive at low cuts for a different reason – the likely one, not tested separately here: with few bands kept per bin, a small change of a bin's σ_v and a small change of its linear bias $b_{1,i}$ move those few bands nearly proportionally, so the per-bin dispersion and the per-bin bias are nearly degenerate, and freeing both per bin lets that degeneracy eat the FoM. As the cut rises the bands separate the two effects and the cost of the bound falls to the 5–11% above.

8.4 Numerical checks

The whitened norm. The checks below measure changes of the data vector in the whitened norm $\|x\| = \sqrt{x^{\mathsf{T}} G x}$ with $G = B^{\mathsf{T}} C^{-1} B$. Here the columns of B are the derivative columns of the base (each the change of the whole data vector per unit change of one parameter), x holds

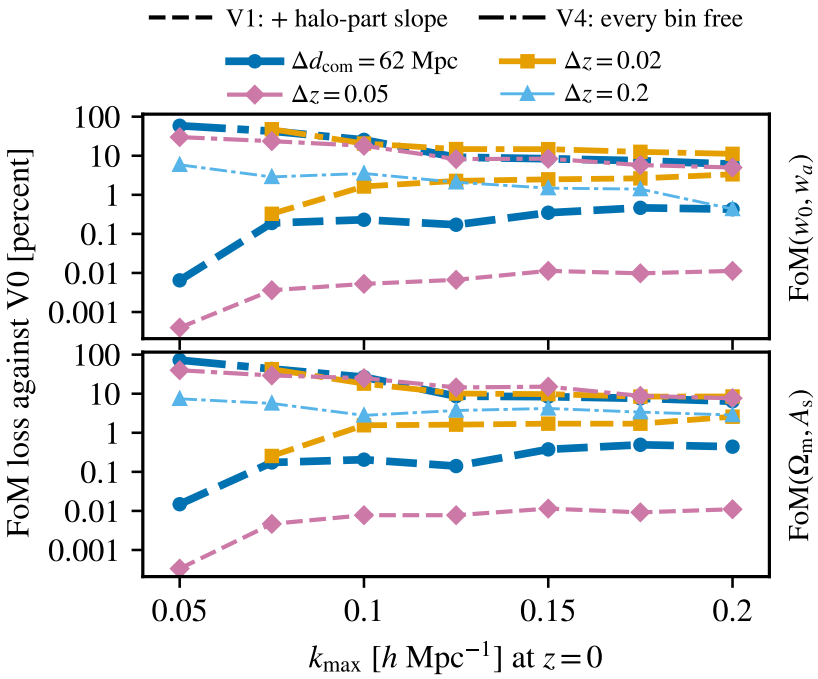


Figure 8: The cost of the per-bin freedom: the loss of FoM(w_0, w_a) (top) and FoM$(\Omega_m, 10^9 A_s)$ (bottom) against today's model V0, in percent, for the slope variant V1 at the baseline width (dashed) and for the bound V4, every bin's dispersion free (dash–dotted). V1 on $\Delta z = 0.2$ is not drawn: its loss stays below 10^{-4} percent. The other slope widths and the residual variants are quoted in the text. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

the coefficients of a change in that base, so Bx is the change of the data vector itself, and C is the data covariance of the bands the shipped mask keeps: $x^{\mathsf{T}}Gx = (Bx)^{\mathsf{T}}C^{-1}(Bx)$ is the χ^2 of that change. "Whitened" means the change is expressed in units of the data's own error bars, every band weighted by how well it is measured: a norm of 1 is a change of the data the size of its one-sigma noise.

Four checks.

- **Group sum.** The sum of a group's per-bin σ_v columns reproduces the group's own σ_v column of the paper's derivative base – an independent calculation – to 0.1–0.5% in the whitened norm: at worst 0.54% ($\Delta z = 0.05$), 0.11% ($\Delta z = 0.02$), 0.16% ($\Delta d_{\rm com} = 62$ Mpc) and 0.15% ($\Delta z = 0.2$).
- **Reproduction of the notebook.** V0 of the study reproduces the Fisher notebook's FoM and its η to 10^{-9} (the study reuses the notebook's own fit cells).
- **Reset.** The model evaluated at the fiducial before and after all the steps agrees exactly (difference 0).
- **Step test.** Per-bin columns from relative steps 0.01 and 0.04 differ from the step-0.02 columns by 3.5–5% ($\Delta z = 0.2$ and $\Delta d_{\rm com} = 62$ Mpc) and 8–10% ($\Delta z = 0.02$ and 0.05)

in the whitened norm, column by column. The whole study was refitted with step 0.01 and with step 0.04 as the main step (the fits only; the caches hold all three steps): every number of Figures 6–8 except V4 is unchanged to the third digit – on $\Delta z = 0.02$ at $k_{\max} = 0.2$ the $T_{\rm lin}$ shift of (w_0, w_a) is 0.3152 / 0.3152 / 0.3151 for steps 0.01 / 0.02 / 0.04, η of V0 under $T_{\rm lin}$ + B is 1.0627 at all three, and the V1 loss is 3.349 / 3.347 / 3.346%. V4's losses move by at most 0.2 percentage points, except $\Delta z = 0.05$ at $k_{\max} = 0.05$ with step 0.04 (30.0 / 30.0 / 36.3%), where few bands are kept.

Two of these numbers deserve a plain-words reading. First, how can a 10% difference between derivative columns leave every shift and FoM unchanged to the third digit? Because the difference lives in the per-bin directions the data barely constrain: those directions' errors are set by the priors, not by the columns' small details, so the columns can wobble there without moving any score. Second, the group-sum check is best at step 0.02 (0.1–0.5%, against 1–4% at 0.01 and 0.04) – and 0.02 is the step of the paper base's own group columns, exactly as expected when the two sides of the comparison share the same stencil.

8.5 What to remember, and an option

- The linear flow's decline across each group is a deterministic mismatch of today's flat-σ_v model, fixed in sign and size by the fiducial cosmology: it moves the (w_0, w_a) fit by up to a third of its error ellipse on the thinnest layout at $k_{\max} = 0.2\,h/\mathrm{Mpc}$.
- One free halo-part slope per group restores most of the error bars' honesty (η from 1.063 to 1.016) at a cost of at most a few percent of either FoM, but absorbs only a third of the $T_{\rm lin}$ shift: the slope $T_{\rm lin}$ implies is about one sigma of the baseline prior, so the prior shrinks the fitted slope partway and a shift of a fifth of the ellipse remains.
- The projected residuals buy almost nothing and cost almost nothing: they carry neither a level nor a slope, and in σ_v units their priors are a few percent.
- Every per-bin σ_v free is the expensive bound, because at low cuts a bin's dispersion and its linear bias are nearly degenerate in the few kept bands.
- Thin bins are the sensitive ones: the blur relocates a share $\sigma_z/(\sqrt{2\pi}\,w_{\rm bin})$ of a bin's galaxies, and the $\Delta z = 0.02$ bins are 37–52 Mpc deep at $z \gtrsim 1$, so that layout responds most and $\Delta z = 0.2$ barely at all.

An option: carry the linear part in the fiducial. The $T_{\rm lin}$ shape is not a nuisance to marginalize: it is fixed by the fiducial cosmology (Section 4.4). A model whose fiducial carries it – $\sigma^2_{v,i} = \sigma^2_{\rm lin}(\bar{z}_i) + \sigma^2_{{\rm HOD},g}$, with one free level per group exactly as today – would remove the $T_{\rm lin}$ shift with no new parameter, hence at no FoM cost to first order, and would leave the slope prior to cover the halo part's own evolution. It changes the fiducial dispersions by up to 10 km/s, so the paper's derivative bases would be out of date and the expensive derivatives would have to be recomputed at the new fiducial. Whether to do that is the project lead's decision (the appendix lists it). A cheaper intermediate needs no new derivatives: center V1's slope prior on the slope $T_{\rm lin}$ implies instead of on zero.

A Open issues

Runs. The derivative columns, the step test, the group-sum check, the accuracy test and the FoM variants are done (Section 8). Still open, and optional: the same per-bin columns on the largest-pair-list bases. The refits with $w \pm 0.05$ are closed without new runs: w enters only as the factor multiplying the halo-part widths (Eq. (14)), so a change of w by 0.05 – 17–35% of w – rescales every halo-part prior by that factor, well inside the 0.5–2.0 slope-width bracket the study already computed.

Decisions for the project lead.

- **The linear-part template.** Whether to build the linear flow's redshift dependence into the fiducial (Section 8.5): it removes the $T_{\rm lin}$ shift with no new parameter, and it makes the paper's derivative bases out of date, so the expensive derivatives must be recomputed at the new fiducial. The cheap intermediate – centering the slope prior on the slope $T_{\rm lin}$ implies – needs no new derivatives.
- **The source forecast's per-bin $\sigma_v(\bar{z})$ figure** is not available now. The priors of Table 5 stand on the literature and the halo model; when the figure turns up, a χ^2 of its points against a flat line per group bounds the real bin-to-bin scatter, and σ, L and the slope widths are refitted in minutes with no new derivatives.
- **Robustness study or fiducial-model candidate?** Whether a per-bin variant that passes the accuracy test is a check on today's model or a candidate to replace it decides whether the summary note changes.
- **Confirm the halo-part priors** of Table 5 (slope widths, A, B, BGS factors) and whether the white term is included.
- **BGS flux-limited** is settled project-wide (the bias drift already comes from the flux limit), fixing its slope width at 2.0; reopen only with that decision.

Open physics items.

- **The ELG halo part is model dependent at the factor-two level** (Section 4.5): the subtraction gives 114–119 km/s against 76–145 km/s from the two published occupations. Fractional priors on $\sigma_{\rm HOD}$ absorb this; an absolute prior would not.
- **The blur's Gaussian shape** against the mixture's heavier tails (Section 4.7) is an open robustness check on the profile, separate from the per-bin amplitude freedom studied here.
- **The Kaiser boundary**: damping with the full one-point σ_v versus the halo part alone (Section 4.7) is a modeling convention the project inherits from the source forecast; the wide paper priors cover it, and no result of this study depends on settling it.
- **Mass-definition conversions** (CompaSO to M_{200c}, Section 4.1) shift $\sigma_{\rm halo}$ by 1–3% and are absorbed into the priors; a precision reanalysis of the halo model would apply them explicitly.

- **The halo-motion part** is linear theory with a several-percent uncertainty of either sign (Section 4.4); it reaches the study only through w, and the $w \pm 0.05$ rescaling argument above bounds its effect inside the computed slope-width bracket.

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